E-Commerce Analysis Business Insights

Project: E-Commerce Orders Performance & Customer Analytics
Tools: Excel, Python + Power BI

This report summarizes the main business insights identified from the cleaned e-commerce dataset and the Power BI Performance dashboards.

1. Executive Summary

METRIC	RESULT
Total Revenue	11.37M
Total Cost	9.15M
Total Profit	2.22M
Profit Margin	19.49%
Total Quantity Sold	92K
Average Review Rating	4.05

2. Revenue by Product Category

Electronics generated the highest sales at approximately 6.3M, followed by Home & Kitchen (1.8M) and Fashion (1.3M). Books recorded the lowest category sales at approximately 0.2M. The difference between the highest and lowest categories is about 6.1M, showing heavy revenue concentration in tech hardware.

CATEGORY	REVENUE
Electronics	6.3M
Home & Kitchen	1.8M
Fashion	1.3M
Sports	0.7M
Beauty	0.5M
Groceries	0.4M
Toys	0.3M
Books	0.2M

3. Revenue by Customer Segment

Returning customers contributed the highest revenue at approximately 4.6M, followed by New customers (3.0M) and Loyal customers (2.7M). Premium tier recorded the lowest sales volume at approximately 1.1M. Returning and loyal cohorts together generate over 64% of total sales.

CUSTOMER SEGMENT	REVENUE
Returning	4.6M
New	3.0M
Loyal	2.7M
Premium	1.1M

4. Customer Metrics by Device & Gender

Desktop represents the largest share of customer visits at 50.09% (8K customers), followed by Mobile at 36.55% (6K) and Tablet at 13.36% (2K). Sales amount by gender is evenly distributed between Female (48.16%) and Male (48.04%), with Other accounting for 3.8%.

DEVICE TYPE	CUSTOMERS	SHARE (%)
Desktop	8K	50.09%
Mobile	6K	36.55%
Tablet	2K	13.36%

5. Revenue by Marketing Traffic Source

Social Media drove the highest sales volume at approximately 1.98M, closely followed by Direct (1.95M) and Email marketing (1.90M). Organic Search generated 1.88M, while Paid Ads and Referral each generated 1.83M. The channels are well balanced with only a 0.15M gap between the highest and lowest sources.

TRAFFIC SOURCE	REVENUE
Social Media	1.98M
Direct	1.95M
Email	1.90M
Organic Search	1.88M
Paid Ads	1.83M
Referral	1.83M

6. Marketing Efficiency & Discounting

The platform recorded an Average Order Value (AOV) of $379.00 with a Coupon Used Rate of 68.10% and an Average Discount of 12.62%. On revenue by device, Mobile accounts for the vast majority of transactional volume at 7.1M (62.41%), compared to Desktop at 3.4M (29.9%) and Tablet at 0.87M (7.69%).

MARKETING METRIC	RESULT
Average Order Value (AOV)	$379.00
Coupon Used Rate	68.10%
Average Discount Percent	12.62%
Mobile Revenue Share	62.41% (7.1M)

7. Fulfillment & Order Status

Standard delivery generated the highest fulfillment revenue at 6.2M, followed by Express shipping (3.0M), while Pickup and Same Day generated 1.1M each. Approximately 80.28% (7.4M) of total orders were delivered successfully, 12.16% (1.12M) were returned, and 7.56% (0.7M) were cancelled.

SHIPPING METHOD	REVENUE
Standard	6.2M
Express	3.0M
Pickup	1.1M
Same Day	1.1M

8. Order Returns by Product Category

Overall Return Rate stands at 10.11% with an average delivery duration of 5 days. Fashion recorded the highest number of returned orders at 845, followed by Groceries (475) and Electronics (432). Toys recorded the lowest return count at 182 units.

CATEGORY	RETURNED ORDERS
Fashion	845
Groceries	475
Electronics	432
Home & Kitchen	363
Beauty	261
Sports	239
Books	236
Toys	182